ZHENYU XIAO

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PERSONAL SUMMARY

Treasury ALM professional building AI-native financial analytics, arbitrage systems and research tools.

PROFESSIONAL EXPERIENCE

American Express | New York, NY

Manager, Asset and Liability Management | Mar 2024–Present

Analyze 12-/24-month Earnings at Risk (EaR) and Economic Value of Equity (EVE) across a balance sheet with approximately $300B in assets.

ALM Engine: projects interest income and economic value under interest-rate shocks.

- Built an in-house calculation engine alternative to Moody's RiskConfidence with AI agents, covering all team risk scenarios and targeting $1M+ in annual software savings.
- Delivered a self-service app in six weeks, replacing legacy benchmarking for 10 team members.
- Designed multi-agent task allocation, persistent context, automated tests and review loops; translated cash-flow rules into executable acceptance criteria.
- Validated numerical parity with RiskConfidence: <0.1% NII/EVE deviation by contract type across six months.
- Cut runtime for 56 risk scenarios by 75%, from 8 hours to 2 hours.

ALM Analytics: automates financial inputs, interest-rate analysis and software guidance.

- Engineered an object-oriented Python library for yield-curve modeling and scenario analysis across 54 curves, six shocks and a 30-year horizon.
- Built AI-assisted financial ETL pipelines for 10+ monthly inputs, cutting preparation from 5 hours to 10 minutes.
- Validated 100K+ balance-sheet records and 24-month forecasts; analyzed curve, APR-cap and deposit-beta effects on earnings and economic-value risk.
- Configured a retrieval-augmented Microsoft Copilot custom agent over ~10 technical manuals for domain-specific financial-software Q&A.

SELECTED PROJECTS

Polymarket US: AI-native development of event-driven arbitrage infrastructure.

Live personal-capital trading | Aug 2026–Present

- Built low-latency arbitrage infrastructure with Codex agents, deploying multi-leg and cross-exchange strategies in live markets.
- Generated $15K+ account net PnL on $1.5K contributed capital (>1,000% return), Aug 3–Oct 10; includes liquidity rewards.
- Engineered deterministic payoff models and outcome-space enumeration to identify arbitrage across logically linked prediction markets.
- Optimized execution with WebSocket feeds, prewarmed connections and asynchronous orders; reconciled fills and uncertain order states via private events and REST.

Trading Bot: an AI-assisted crypto strategy research and backtesting prototype.

- Built a FastAPI/LLM pipeline for evidence retrieval, structured strategy cards and candidate ranking.
- Built BTC/ETH walk-forward backtests with transaction costs, Sharpe and drawdown analysis.

SKILLS

Technical: Python, SQL, Git, FastAPI, REST APIs, WebSockets; LLM APIs, Codex, GitHub Copilot, RAG/custom agents, multi-agent orchestration, context management.

Finance: Yield-curve and cash-flow modeling, interest-rate risk, arbitrage, backtesting, scenario analysis, Bloomberg.

EDUCATION

Columbia University | M.S., Operations Research | Jan 2024

City University of Hong Kong | Bachelor of Finance | Jun 2022